

Machine Learning Project

Corporate Bankruptcy Prediction Using Financial Ratios

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GitHub repository:

<https://github.com/motheeric/Machine-Learning-project>

Abstract

Predicting corporate bankruptcy is an important problem in financial risk management. Early detection of financially distressed companies can help banks, investors, and regulators reduce losses and take better decisions. In this project, we apply machine learning techniques to predict bankruptcy using financial ratios. We follow a complete machine learning pipeline, from data exploration to model evaluation, with a particular focus on class imbalance and model generalisation.

1 Introduction and Motivation

Corporate bankruptcy can have significant economic consequences for companies, investors, and the financial system as a whole. Being able to identify firms at risk of bankruptcy before default occurs is therefore a key objective in credit risk assessment and investment analysis. Traditional approaches rely on expert judgement and financial rules, but they can be limited when dealing with large and complex datasets.

With the increasing availability of financial data, machine learning methods offer powerful tools to complement standard financial analysis. These methods are able to capture non-linear relationships between financial indicators and provide scalable decision support. The objective of this project is to explore how machine learning models can be used to predict corporate bankruptcy in a realistic setting and try to help dying companies.

2 Problem Formulation

The task is formulated as a supervised binary classification problem. Each observation corresponds to a company described by a vector of financial ratios, and the target variable indicates whether the company went bankrupt or not it is 0 or 1.

Let $X \in \mathbb{R}^{n \times p}$ be the matrix of input features and $y \in \{0, 1\}^n$ the target variable, where $y = 1$ represents a bankrupt company. The goal is to learn a function f such that $\hat{y} = f(X)$ generalizes well to unseen companies. Due to the imbalanced nature of the dataset, accuracy alone is not sufficient to evaluate model performance that is why we did multiples models to estimate wether or not bankruptcy will happen.

3 Dataset Description

The dataset used in this project is the *Company Bankruptcy Prediction* dataset, which contains financial ratios extracted from company balance sheets and income statements. It includes 6,819 observations and 95 numerical features, as well as a binary target variable named **Bankrupt?**.

The features describe different aspects of a firm's financial health, such as profitability, liquidity, solvency, efficiency, and cash flow. This dataset reflects real-world conditions, including noisy measurements and a strong imbalance between bankrupt and non-bankrupt companies.

4 Exploratory Data Analysis

4.1 Class Balance

An initial analysis of the target variable shows a strong class imbalance, with bankrupt companies representing only a small fraction of the dataset. This imbalance is a challenge, as a naive classifier predicting only non-bankruptcy would still achieve high accuracy.

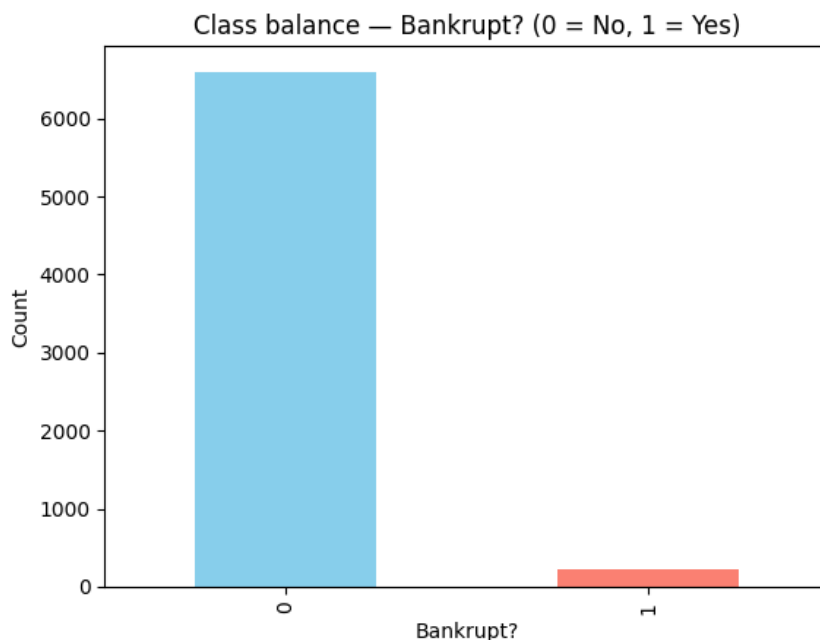


Figure 1: Distribution of the target variable (**Bankrupt?**), highlighting the strong class imbalance.

4.2 Correlation Analysis

Many financial ratios measure related economic quantities, which results in strong correlations between features. The correlation matrix highlights several blocks of highly correlated variables, suggesting that the variables are almost identical in the dataset and motivating dimensionality reduction.

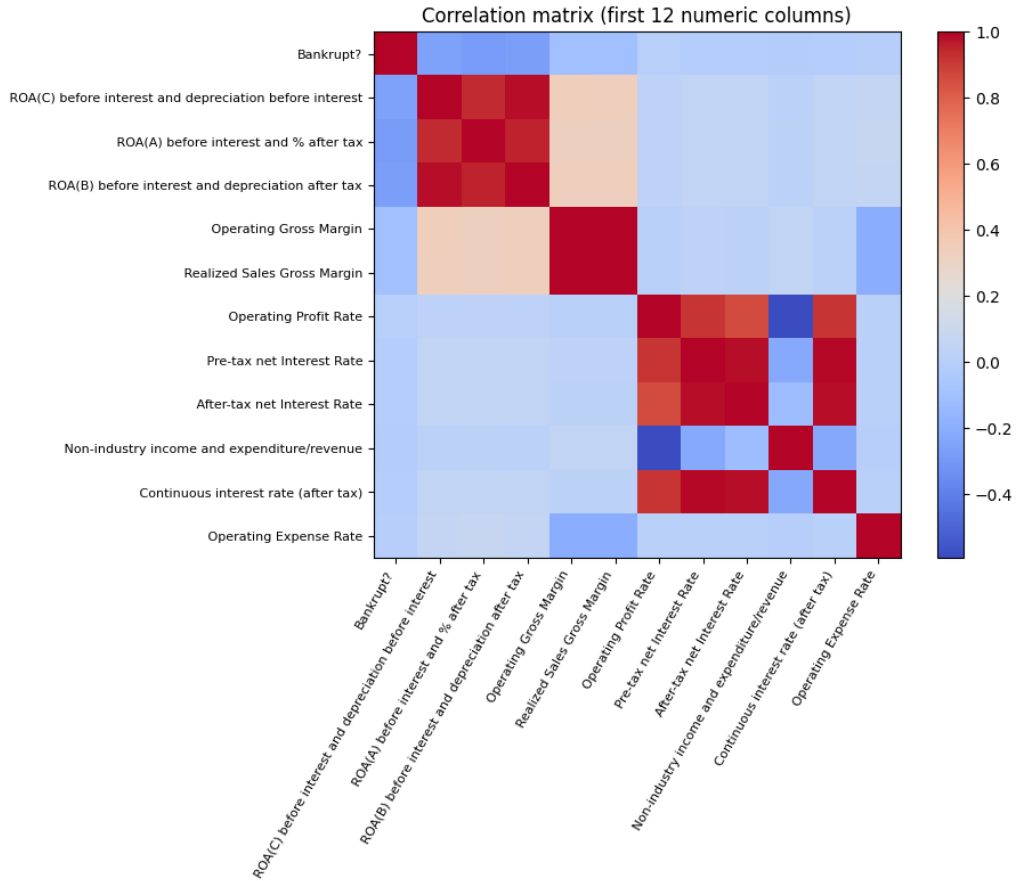


Figure 2: Correlation matrix of financial ratios showing groups of strongly correlated features.

5 Methodology and Preprocessing

Before training the models, several preprocessing steps are applied. The dataset is split into training and testing sets using stratification to preserve the proportion of bankrupt companies. Feature scaling is performed to standardize the variables, which is especially important for linear models such as Logistic Regression.

To better understand the structure of the data, Principal Component Analysis (PCA) is applied after scaling. The cumulative explained variance shows that a reduced number of components captures most of the information contained in the original features.

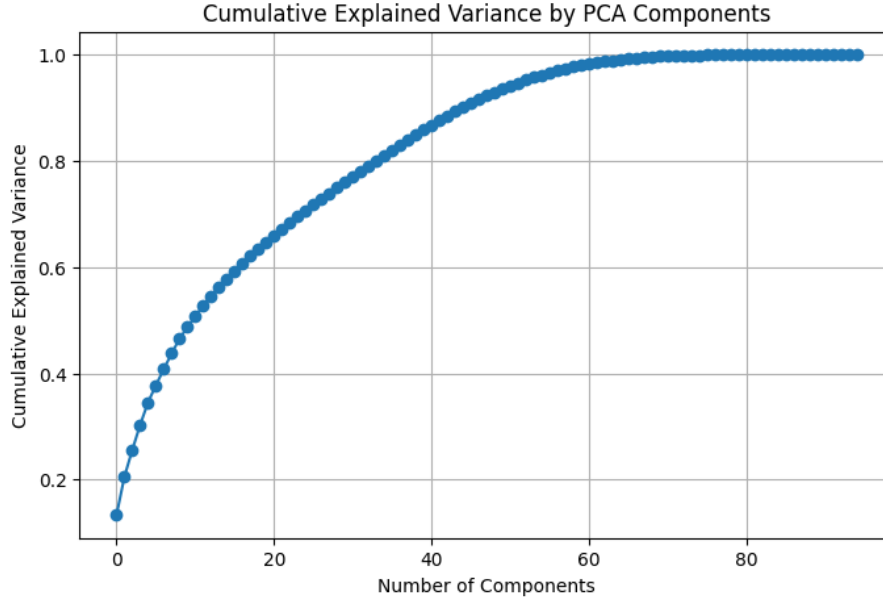


Figure 3: Cumulative explained variance of PCA components, illustrating feature redundancy.

6 Models

6.1 Logistic Regression

Logistic Regression is used as a baseline model due to its simplicity and interpretability. weights are introduced to reduce the impact of class imbalance. This model provides a reference point for evaluating more complex approaches after.

6.2 Random Forest

Random Forest is an ensemble learning method capable of capturing non-linear relationships between financial datas. It is particularly well suited for tabular financial data and often achieves strong predictive performance without heavy features.

7 Evaluation and Model Analysis

To evaluate the models, we focus on metrics adapted to imbalanced classification, such as ROC-AUC, recall, and F1-score. Learning curves are used to analyse the generalization behavior of the models and to detect potential underfitting or overfitting.

For Logistic Regression, the learning curve suggests limited performance due to model simplicity and strong feature correlations.

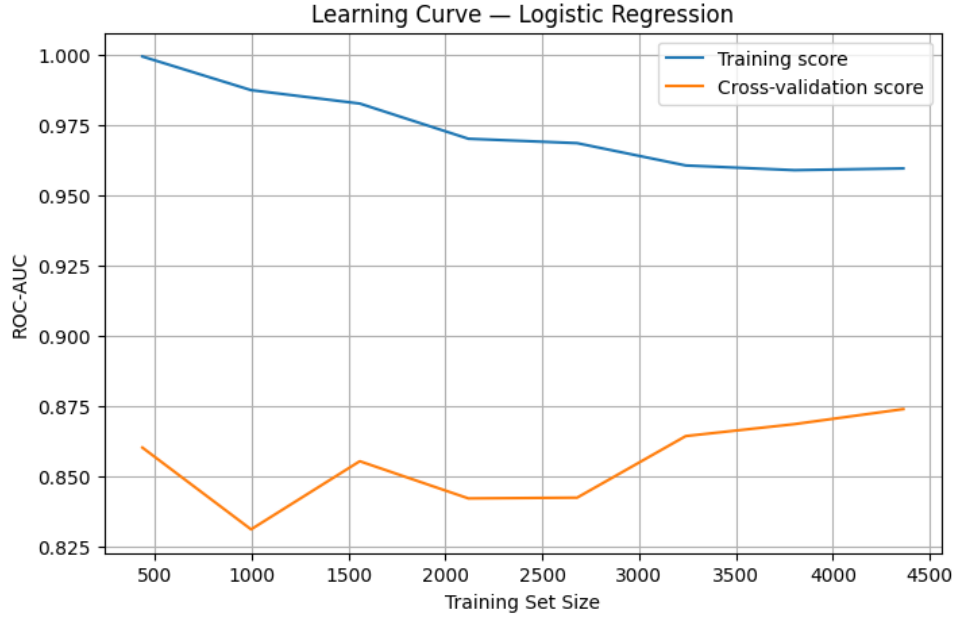


Figure 4: Learning curve for Logistic Regression, showing limited generalization capacity.

In contrast, the Random Forest model achieves higher validation performance and shows better generalisation, indicating a stronger ability to capture complex patterns in the data.

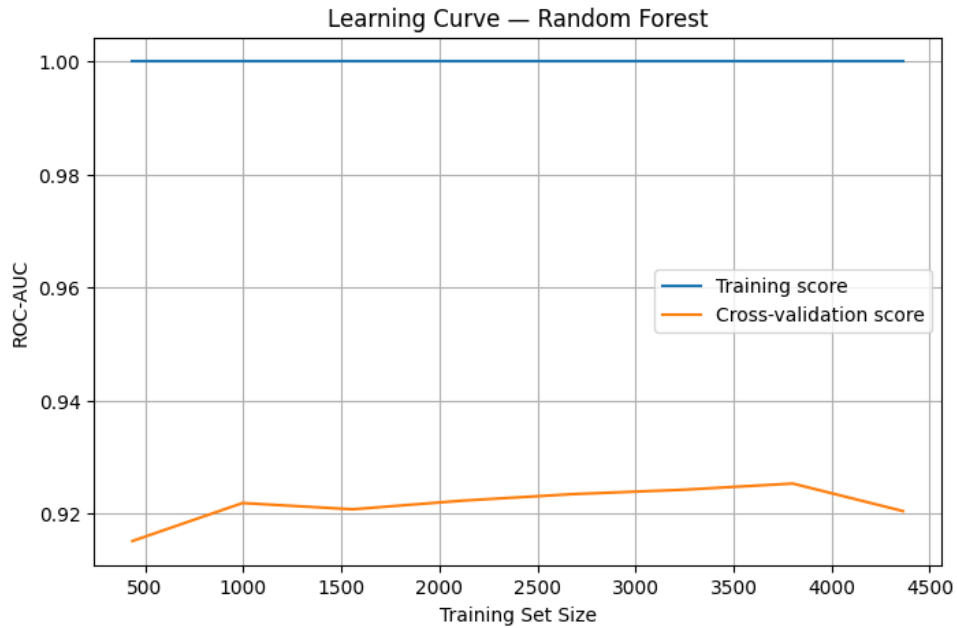


Figure 5: Learning curve for Random Forest, highlighting improved performance compared to the baseline.

8 Limitations and Discussion

Despite encouraging results, several limitations remain. The model relies only on accounting ratios and does not include macroeconomic or temporal information, which may also influence bankruptcy risk. In addition, more complex models tend to be less interpretable, which can be a constraint in financial decision-making contexts.

9 Advanced Models, Ensembling Strategies, and Hyperparameter Optimization

Beyond the baseline models, we evaluated several advanced algorithms that are known to perform well on structured tabular data, including gradient boosting methods, neural networks, and ensemble strategies. These models are particularly relevant in settings with complex feature interactions and class imbalance, such as our project.

9.1 Hyperparameter Tuning with GridSearchCV

Hyperparameters were optimized using `GridSearchCV` with cross-validation on the training set. Given the strong class imbalance, the cross-validation score was used to compare models in a consistent and reproducible way.

The Logistic Regression baseline achieved a relatively low cross-validation score (**CV score = 0.39**), indicating limited capacity to capture the complexity of the data.

In contrast, **LightGBM achieved the best cross-validation performance** among all tested models, with a **CV score of 0.47**.

9.2 Gradient Boosting Models

9.2.1 LightGBM (Final Model)

LightGBM is a gradient boosting framework optimized for efficiency and performance on large and high-dimensional datasets. In our experiments, LightGBM clearly outperformed other models in terms of cross-validation score and overall robustness.

On the test set, LightGBM achieved the following F1-scores:

- Class 0 (non-bankrupt): **F1-score = 0.98**
- Class 1 (bankrupt): **F1-score = 0.50**

These results highlight the strong ability of LightGBM to correctly classify the majority class while maintaining reasonable performance on the minority class (bankrupt), which is critical in a highly imbalanced setting.

9.2.2 XGBoost

XGBoost is another widely used gradient boosting implementation with strong regularization mechanisms. In our experiments, XGBoost achieved performance comparable to LightGBM on the majority class, but did not surpass it overall.

The test set F1-scores for XGBoost are:

- Class 0 (non-bankrupt): F1-score = 0.98
- Class 1 (bankrupt): F1-score = 0.50

Although it was great, XGBoost did not provide a clear improvement over LightGBM and was therefore not selected as the final model.

9.2.3 Gradient Boosting (sklearn)

We also evaluated a classical Gradient Boosting classifier. While it performed very well on the majority class, its performance on the minority class was weaker compared to LightGBM and XGBoost.

- Class 0 (non-bankrupt): F1-score = 0.98

- Class 1 (bankrupt): F1-score = 0.43

This confirms that more advanced boosting frameworks are better suited for this task.

9.3 Ensemble Methods

9.3.1 Soft Voting Classifier

To improve robustness, we implemented a **soft Voting Classifier**, which combines the predicted probabilities of several base models. Soft voting is particularly appropriate in imbalanced classification settings, as it boosts probabilistic information rather than hard class labels.

9.3.2 Stacking Classifier

We also tested a Stacking Classifier, where multiple base learners are combined through a meta-model. While stacking can outperform individual models when base learners are highly complementary, the performance gains observed here were limited..

9.4 Final Model Selection

Based on cross-validation results, test performance, and model robustness, **LightGBM was selected as the final model**. It achieved the highest CV score (**0.47**) and strong F1-scores on both classes, making it the most suitable model for corporate bankruptcy prediction in this project.

10 Conclusion

This project highlighted several practical challenges commonly encountered in real-world machine learning applications, particularly in financial risk prediction. One of the main difficulties was the strong class imbalance in the dataset, which made accuracy an unsuitable evaluation metric and required the use of adapted measures such as F1-score and recall for the minority class.

Another important challenge was the high level of correlation between financial ratios. Many variables had overlapping information, which increases the risk of multicollinearity and can negatively impact certain models. This motivated the use of dimensionality reduction techniques, as well as the preference for models correlated features.

Model selection and hyperparameter also proved to be hard. Simple baseline models provided useful reference points but were often limited in their ability to capture complex patterns in the data. More advanced models improved performance but introduced additional risks such as overfitting, which required careful analysis using learning curves and cross-validation.

Finally, this project emphasized the importance of adopting a rigorous and structured machine learning workflow. From data analysis to model diagnostics, each step played a key role in building reliable and interpretable results. These methodological choices reflect realistic constraints faced in financial machine learning projects and provide valuable insight for future work in risk modeling and decision support.